Beaver engineering creates intense but bidirectional dissolved organic carbon hotspots in streams

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Abstract

Despite recolonization across Europe, the effects of beavers on freshwater dissolved organic carbon (DOC) concentrations remain understudied. DOC influences microbial activity and energy flow in aquatic food webs. Understanding how and when beaver-engineering alters DOC is therefore critical for predicting its ecosystem consequences. Using a national-scale dataset we quantified the magnitude and drivers of DOC concentration changes associated with beaver-engineering. We found that beaver-engineering exerted negligible influences on

DOC in about 76% of stream samplings, but also promoted DOC source (12–16%) and sink (9–12%) responses. Dams modified stream hydrology, which influenced DOC indirectly via changes in primary producer abundance. Compared to floodplain-scale DOC variation, beaver-engineering amplified DOC concentration changes 140 times in summer and 32 times in winter. These findings are crucial as they enable stakeholders to predict whether beaver-engineering will increase or decrease DOC concentrations, guiding decisions on where and when beaver presence will affect water quality and ecosystem functioning.

Keywords: Castor, dam, freshwater, DOC, causal inference, abundance, primary producers, hydrology

1 Introduction

Ecosystem engineers are organisms that modify the physical structure of their environment and thereby control the availability of resources to other organisms [1]. They can profoundly alter ecosystem function without necessarily shifting ecosystem properties in a consistent direction across landscapes: by restructuring habitat, hydrology and resource availability, they create localized control points where biogeochemical processing becomes disproportionately intense [2]. Beavers (*Castor fiber* L. and *C. canadensis* Kuhl), whose populations are recovering across Europe and North America with assistance from rewilding and restoration, offer a powerful and increasingly widespread example of such engineering: their dams profoundly influence riparian system hydrology, ecosystems and biogeochemistry [3, 4].

Beaver valley-floor modifications are known to reshape organic matter (OM) and organic carbon (OC) dynamics across river corridors. Beaver ponds and meadows trap fine sediment and particulate OC, store large amounts of carbon in pond sediments and floodplain soils, and raise water tables that slow OC mineralization [5–8]; their abandonment and drainage can release this stored carbon [6, 9]. By creating ponds and wetlands, beavers also reduce water velocity, increase lateral river–floodplain connectivity, and favour the growth of macrophytes and aquatic primary productivity by opening the canopy [10–12]. These modifications can promote enrichment of the dissolved fraction of OC (dissolved organic carbon, DOC) through aquatic production and exudation, vegetation turnover and litter inputs, and inundation and leaching of organic-rich soils [10, 13, 14]; however, they may also promote DOC depletion via longer processing times, microbial mineralization, photochemical processing, retention, and sediment and subsurface pathways [11, 15]. Thus, there is no strong theoretical reason to expect beaver ponds consistently to be DOC sources or always to be DOC sinks; indeed, syntheses and reviews find that beaver ponds increase DOC in the majority of cases, but not consistently [3, 4].

DOC itself is key to the carbon (C) cycle and plays an important role in safeguarding freshwater biodiversity and maintaining water quality [16]. It links terrestrial and aquatic carbon pathways and, as a major energy source for microorganisms, provides the substrate for heterotrophic metabolism, influencing microbial respiration, nutrient cycling and overall ecosystem functioning in streams and rivers [17–19]. Elevated DOC can lead to hypoxia or other unfavourable conditions for aquatic life [20], and DOC contributes to the formation of disinfection byproducts in drinking water treatment, making its monitoring and management important for human health [21]. DOC is therefore a critical pathway by which the physical engineering of beavers can be transmitted into food webs and water quality.

Yet, despite the potential impact of beavers on DOC transformations, the effects of beaver-engineering on DOC have been shown to vary widely depending on site-specific factors such as vegetation, soil, hydrology and climate [3, 4], and DOC dynamics have been studied mostly at individual sites [13–15]. It thus remains unclear whether the inconsistent responses reported among sites represent noise around a common average effect, or whether context-dependent positive and negative responses are themselves a characteristic feature of beaver ecosystem engineering. No large-scale analyses have been available so far to examine how beaver-induced hydrological and biogeochemical modifications influence the timing and magnitude of DOC changes across diverse landscapes. Such an analysis must account for the drivers that control stream DOC independently of beavers: land use within the catchment, the diversity and abundance of primary producers, climatic conditions, river–floodplain connectivity and stream hydrology [16, 22]. DOC concentrations vary seasonally, peaking in summer due to higher contributions of lateral and in-stream inputs from aquatic vegetation and lower water flow [23]; forested catchments exhibit higher DOC concentrations due to more abundant OM input [24]; land-use change such as urbanization and agriculture can alter the quantity and quality of DOC [25]; and the combined effect of river–floodplain connectivity and waterflow regimes can cause rapid fluctuations in DOC concentrations [24, 26].

In this study, we used a national-scale dataset to assess how beaver-engineering and its effect on stream hydrology and primary producer abundance influenced net changes in DOC concentrations (ΔDOC). We proposed the beaver amplification factor (M) to compare how strongly beaver-engineering amplifies DOC concentrations relative to background floodplain DOC changes across the landscape. This approach allowed us to directly contrast beaver-mediated DOC changes with the seasonal floodplain-scale variability in DOC and quantify the magnitude of the beaver-mediated DOC change relative to the floodplain-scale one. We posed the following research questions:

1. *Direction*: does beaver-engineering consistently increase or decrease DOC concentrations downstream compared to upstream?
2. *Drivers*: which abiotic and biotic drivers set the direction of the DOC change in beaver-engineered streams?
3. *Magnitude*: irrespective of its direction, how large is the beaver-mediated DOC change per unit area relative to the seasonal, floodplain-scale DOC variation of the same stream?

We hypothesized that beaver-engineering influences DOC concentrations indirectly by increasing water residence time and enhancing aquatic primary productivity. We expected that this effect is stronger in summer than in winter due to seasonally increased biological activity. Because our study directly compares beaver-mediated DOC changes with background floodplain-scale DOC variation, it provides the evidence base for anticipating where and when beaver presence will affect DOC concentrations, water quality and ecosystem functioning. This work is therefore essential for identifying which drivers shape DOC responses to beaver-engineering and for understanding the implications of beaver-engineering for water quality and ecosystem functioning across landscapes.

2 Results

2.1 Beaver-engineering effects on DOC

Beaver-engineering did not change DOC concentrations in a consistent direction. The average ΔDOC was 0.14 mg/L in summer (95% Bayesian credible interval [BCI]: 0.05 ; 0.22 mg/L; $P(\Delta DOC > 0) > 0.99$) and 0.00 mg/L in winter (95% BCI: -0.06 ; 0.07 mg/L), i.e., a small net DOC release in summer and none in winter, both entirely within the ROPE of $\pm$0.5 mg/L. Around these averages, however, individual responses were highly variable and heavy-tailed (Figure 1a): for a single sampling of a beaver-engineered stream, the posterior predictive probability of a negligible ΔDOC was 75% in summer and 77% in winter, of a DOC source response 16% in summer and 12% in winter, and of a DOC sink response 9% in summer and 12% in winter (95% posterior predictive intervals: -1.9 ; 2.3 mg/L in summer, -2.1 ; 2.1 mg/L in winter). Beaver-engineered streams are therefore about as likely to increase as to decrease DOC by an ecologically relevant amount, with a slight prevalence of source responses in summer, and the heavy tails indicate that occasional large changes occur in both directions. Because ΔDOC is bidirectional, a null average does not imply the absence of an effect: it conceals the magnitude of the change, which we quantify below with the beaver amplification factor M (third research question).

2.2 Abiotic and biotic drivers of DOC in beaver-engineered systems

We found evidence that aquatic primary producers and soil litter cover positively influence ΔDOC (Figure 1b). Our model indicated that beaver-engineering had a positive influence on ΔDOC via an indirect path that is defined as: number of dams $\rightarrow$ water residence time $\rightarrow$ phytoplankton abundance $\rightarrow$ ΔDOC (Figure 1b). Solar radiation also had a positive influence on ΔDOC via an indirect path macrophytes & phytoplankton $\rightarrow$ ΔDOC (Figure 1b). We found that the stream channel gradient had a negative influence on ΔDOC by decreasing water residence time. Notably, upstream DOC concentrations exerted a negative influence on downstream DOC concentrations (Figure 1b). By contrast, we found no clear evidence that maximum dam height affected water residence time or that water residence time directly altered DOC concentrations (Figure 1b). Tracing all causal paths to obtain the total effect of each exogenous driver on ΔDOC (Appendix S1: Figure S7) confirmed that the direction of the DOC change is set chiefly by the external context of the pond: upstream DOC concentration had the largest total effect (standardized effect -0.10; 90% BCI: -0.16

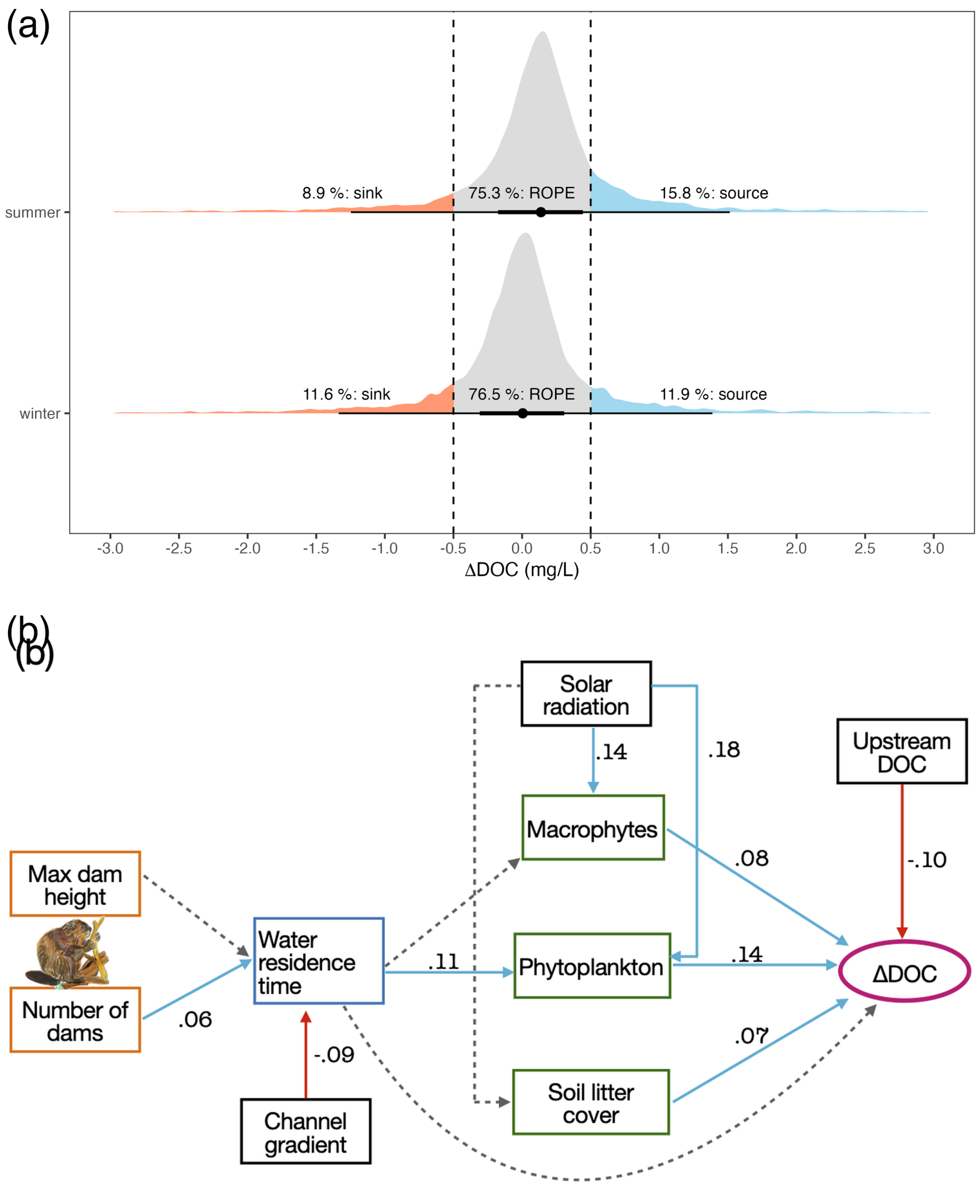

Fig. 1 a) Posterior predictive distribution of ΔDOC for a single sampling of a beaver-engineered stream, in summer and winter (measurement-error model on raw upstream and downstream DOC concentrations). The orange shaded area represents the posterior predictive probability that a single sampling shows a DOC sink response; the light blue shaded area the probability of a DOC source response. The grey shaded area represents the Region Of Practical Equivalence (ROPE) and it defines an interval around zero where effects are too small to be of practical importance. The central dot indicates the median of the distribution, while the thick and thin horizontal lines represent the 66% and 95% intervals. b) Drivers of beaver-mediated DOC change. Solid edges (red = negative, blue = positive) represent 90% probability evidence of causal effects; dashed edges represent inconclusive evidence. Standardized average effects are shown. Full posterior parameter distributions in Appendix S1: Figure S5. Beaver illustration credits to © Sarah Edmonds Illustration.

; -0.05), followed by solar radiation acting through the primary producers (+0.04; 90% BCI: 0.01 ; 0.07), whereas the total effects of the number of dams (+0.003) and maximum dam height (+0.001), which operate only through water residence time, were small. An alternative SCM including lateral stream–wetland connectivity as an additional driver of water residence time yielded path estimates consistent with the main model and no evidence of a connectivity effect on ΔDOC (Appendix S1: Figures S8–S9).

2.3 Magnitude of the beaver-mediated DOC change

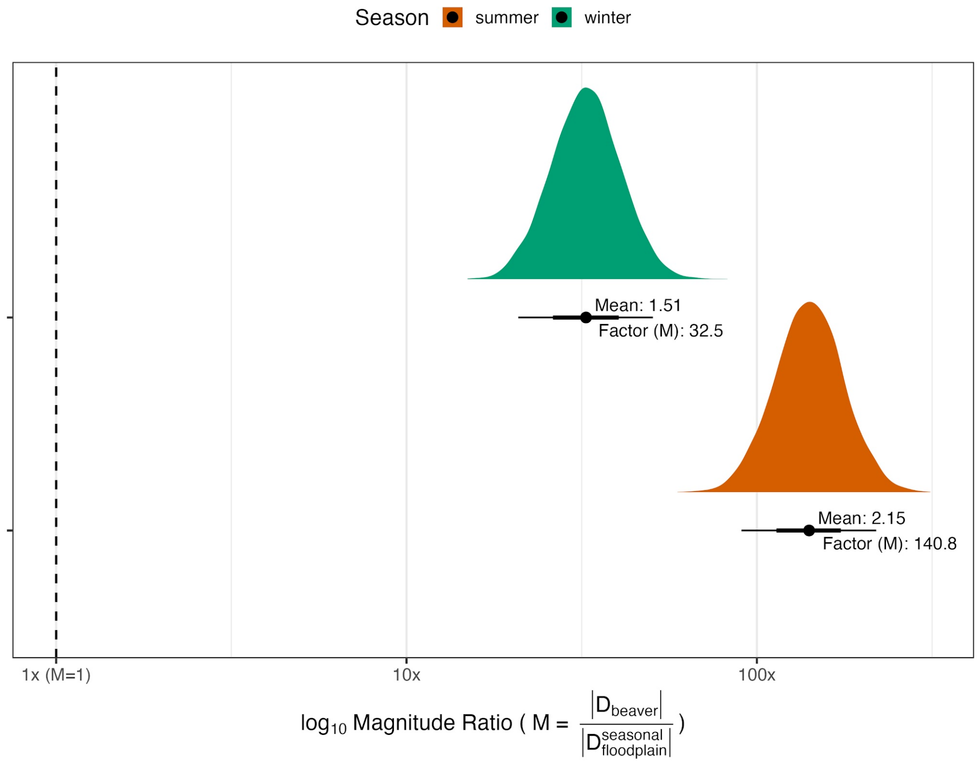

Fig. 2 Posterior distribution of the beaver amplification factor M, comparing the beaver mediated change in DOC to the seasonal floodplain-scale DOC change per unit area. The vertical dashed line at 0 represents the line of equivalence, where the magnitudes are equal. The central dot marks the posterior mean. The thick inner segment shows the 66% credible interval, and the thinner outer segment shows the 95% CI. Green represents winter, and orange represents summer.

Irrespective of its direction, the magnitude of the beaver-mediated DOC change per unit area was on average 140 times (summer) and 32 times (winter) the magnitude of the seasonal floodplain-scale DOC change of the same stream (Figure 2). Since the entire 95% BCIs of both seasons lie to the right of the equivalence line (M = 1), this provides strong evidence that beaver-engineered reaches are hotspots of DOC change in both seasons, and more strongly so in summer, when primary producers are most active.

3 Discussion

We found that beaver engineering does not push stream DOC in a consistent direction: across 176 streams the average change was null, yet roughly one in four samplings of a beaver-engineered reach showed an ecologically relevant DOC change, about as often a source as a sink response (with a slight prevalence of source responses in summer), and the magnitude of that change per unit area exceeded the seasonal floodplain-scale DOC variation by one to two orders of magnitude. Beaver ponds thus act as localized, intense and bidirectional DOC control points [3, 5]: whether a pond exports or retains DOC at a given moment is governed by local hydrology and primary producers rather than being a fixed property of the site. This context-dependence helps to reconcile the conflicting results of previous site-specific studies, which reported beaver ponds as either DOC sources or sinks [3, 4]. We also found that beaver-engineering influenced DOC concentrations indirectly by increasing water residence time, which, in turn, enhanced aquatic primary productivity. Beaver dams are known to slow water flow and increase light penetration, creating stable habitats for phytoplankton and macrophytes [10, 11]. These primary producers were shown to release organic compounds and detritus and to increase DOC concentrations in dammed streams [13]. Although our estimate of the primary-producers effect on DOC was based on a relatively limited number of samples, the positive influence we detected aligns with previously documented relationships between aquatic primary productivity and DOC in lotic ecosystems [16]. Similar indirect effects between water flow alteration, aquatic primary productivity, and DOC were observed in other temperate and boreal streams [3, 4]. Tracing the total effects through the SCM puts this pathway in perspective: the overall influence of dam number and height on ΔDOC via water residence time was small compared with that of upstream DOC load and solar radiation (Appendix S1: Figure S7). Beaver dams thus create the conditions – ponded water, longer residence time, habitat for primary producers – under which DOC can be rapidly produced or consumed, but whether a pond exports or retains DOC at a given moment is decided mainly by what enters it (DOC load) and by the energy available to its primary producers (light, season).

We hypothesised that the influence of beaver-engineering on DOC concentrations would be stronger in summer than in winter, owing to higher primary productivity and warmer water temperatures during the growing season. Our results supported this expectation: the magnitude of the beaver-mediated DOC change per unit area was roughly 140-fold in summer and 32-fold in winter that of the background floodplain-scale DOC change (Figure 2). Seasonally intensified effects of beaver-engineering on DOC mirror patterns described for nutrient cycling in other aquatic ecosystems [3, 4]. The larger magnitude of change in summer likely reflects elevated primary productivity and accelerated microbial turnover [19, 23], both of which are driven by increased solar radiation and warmer temperatures [18]. Taken together, these patterns indicate that seasonality intensifies the effects of beaver-engineering on DOC dynamics by intensifying biologically mediated C fluxes.

Beaver-mediated impacts on DOC concentrations have major implications for aquatic biogeochemistry and food-web dynamics. Elevated DOC concentrations exported during summer can enhance microbial respiration and redirect energy flow from autotrophic to heterotrophic pathways [20, 27]. Such shifts can reduce primary production and alter consumer communities by favouring detritus-based food webs [16]. In winter, DOC retention and sediment C storage may support benthic microbial processes that stabilize nutrient availability and sustain overwintering invertebrates [17, 21]. In low-DOC ecosystems, higher beaver-induced DOC concentrations could challenge drinking-water quality, whereas in high-DOC systems, ponds may instead function as C sinks that enhance habitat complexity and energy support [3, 25].

Quantifying the magnitude of the DOC change also revealed how and when beaver-engineering influenced DOC dynamics in our study. Previous studies have demonstrated spatial and temporal variability in beaver influence on DOC, but seldom expressed them relative to background seasonal variation [10, 15]. By normalizing beaver-mediated DOC change against seasonal floodplain-scale change, we could provide a comparable metric across climatic and ecological settings [3, 11]. This approach emphasizes that beaver-engineering not only modified in situ DOC concentration but also intensified DOC processing far beyond seasonal floodplain baselines, in either direction.

A null average change could also arise if the upstream–downstream difference merely reflected background spatial variability of DOC along a stream. Two observations argue against this interpretation. First, we propagated the analytical measurement error of each DOC concentration (0.2 mg L^{-1}) explicitly through the model; the 95% posterior predictive interval of ΔDOC (about -2 to 2 mg L^{-1}) is an order of magnitude wider, so the variability is not measurement noise. Second, the amplification factor uses the seasonal DOC variation of each stream upstream of the beaver reach as an internal reference of background variability, and the beaver-mediated change per unit area exceeded it by one to two orders of magnitude. We did not sample paired reaches without beaver dams, however, and a direct comparison of upstream–downstream DOC differences in un-dammed reaches of similar length remains a valuable test for future work. Likewise, we did not measure vertical (hyporheic) or lateral exchange between the channel and its floodplain, which can influence DOC independently of beaver engineering [3]; our test of lateral channel–wetland connectivity (Appendix S1) found no detectable influence, but it was based on a coarse, three-level classification.

In conclusion, by using a national-scale dataset, this study provides a comprehensive understanding of which beaver-mediated drivers influence DOC concentrations in streams. It also quantifies how much this beaver-mediated influence exceeds the seasonal floodplain-scale DOC variation. Beaver engineering does not make streams DOC sources or DOC sinks; it creates localized, intense control points where DOC changes far more per unit area than along the floodplain, in a direction set by hydrology and primary producers. By identifying the environmental contexts in which these control points export or retain DOC, this work offers a predictive basis for integrating beaver presence into water-quality management and conservation planning.

4 Methods

4.1 Study sites and sampling design

We chose 176 beaver-dammed sites in Switzerland (Figure 3a). We collected water samples at upstream and downstream locations of the beaver ponds. For each stream, we collected one sample in winter (December, January or February 2021/2022) and one sample in summer (June or July 2022). We collected the upstream sample where the water was still flowing before being dammed. Since, in some cases, there was more than one dam and pond, we took the downstream sample after the last beaver dam in each sampled stream where river flow was not anymore impacted by the dam. A few streams comprised more than one separately sampled beaver-dam sequence, each with its own upstream–downstream pair; these pairs are the statistical units of all models. All sites were single-thread, low-order streams with one to several beaver dams and ponds in sequence along the channel (Figure 3a, b); none of them were extensive beaver meadows with multiple anastomosing channels and ponds at different stages of abandonment as described in North America [3, 6]. For each site we additionally classified the lateral connectivity between the channel and adjacent wetlands (no wetland within 50 m: 136 sites; wetland present but not connected: 11; wetland connected to the channel: 13; 22 sites unclassified) and tested its influence in an alternative structural causal model (Appendix S1). Vertical (hyporheic) and lateral exchange with the floodplain were not measured directly, which we acknowledge as a limitation when interpreting the floodplain-scale reference of the amplification factor (below). Our dataset included a broad set of abiotic and biotic variables that are hypothesized to influence DOC concentrations in freshwater ecosystems (Figure 3; the science-based hypothesis motivating each causal path is given in Appendix S1: Table S1).

4.2 Dissolved organic carbon net change

In each stream, we collected water samples and filtered (0.22 μm) them for analysis of DOC concentration with high-temperature combustion with infrared detection of the CO_2 produced (Shimadzu TOC-L CSH analyzer; detection limit 0.5 $mg\ C\ L^{-1}$; error measurement 0.2 $mg\ C\ L^{-1}$). Across all sites, we calculated the net DOC concentration change (ΔDOC; as our primary response variable), by subtracting DOC concentration at downstream locations with the upstream locations.

4.3 Beaver-engineering variables

We defined the number of dams as the integer number of dams that we counted upstream of the downstream location. We defined the dam height (m) for each site as the maximum dam height observed upstream of the downstream location. Site persistence was defined as the number of years that beavers have inhabited the site.

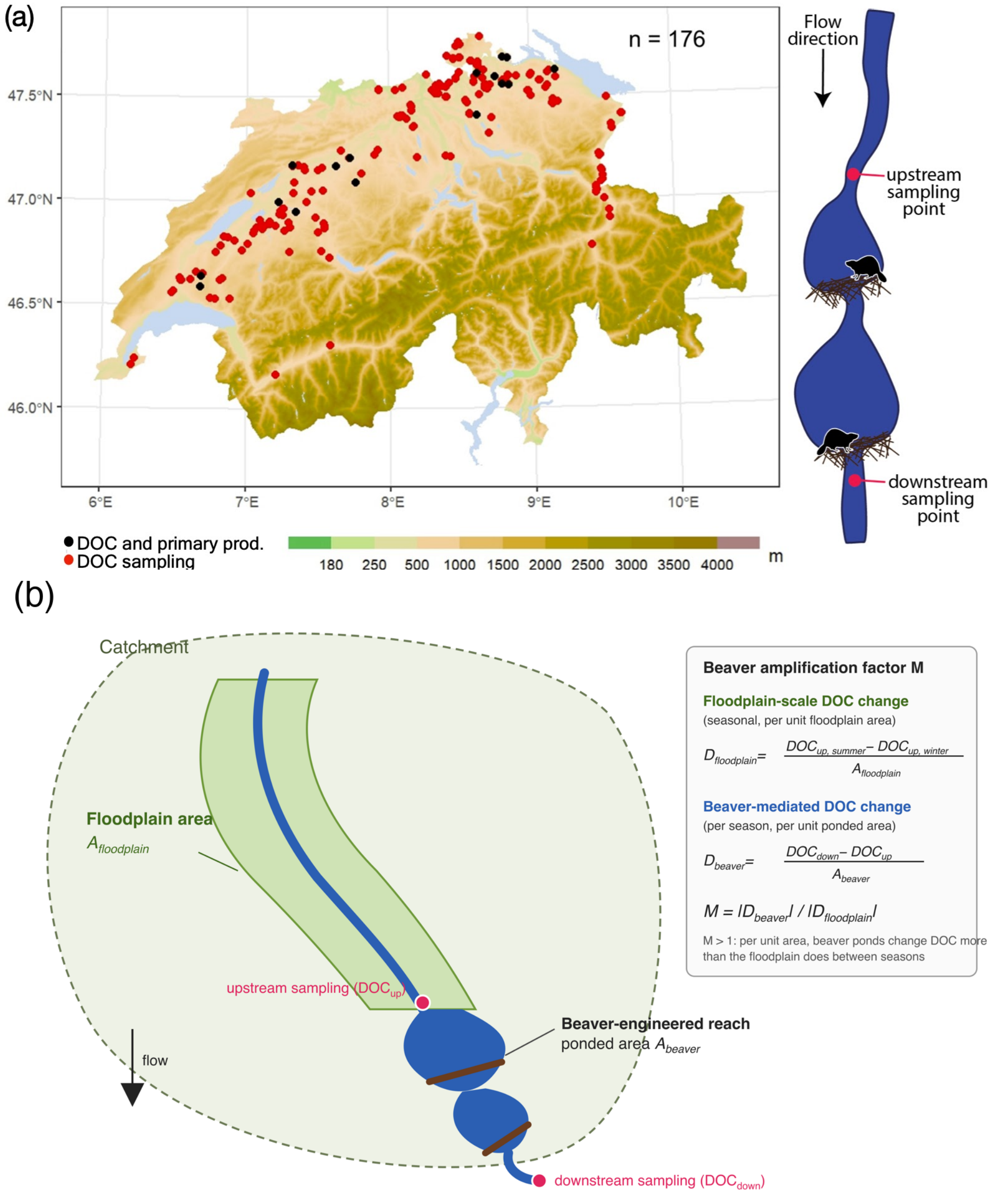

Fig. 3 a) Sampling locations across Switzerland. Red dots (n = 176) denote sites where water samples were collected, black dots (n = 16) denote sites where water samples and primary producer abundance were sampled. The inset shows the sampling design at each site: water was sampled upstream of the first and downstream of the last beaver dam. b) Schematic of the quantities used for the beaver amplification factor M. The beaver-mediated DOC change D_{beaver} is the difference between downstream and upstream DOC concentration divided by the ponded area of the reach (A_{beaver}); the floodplain-scale DOC change $D_{floodplain}$ is the seasonal (summer minus winter) change in upstream DOC concentration divided by the floodplain area of the catchment draining to the site ($A_{floodplain}$). M is the ratio of their absolute values.

4.4 Water residence time

We defined the water residence time (i.e., τ, seconds) using Equation 1:

$$\tau = \frac{V}{Q} \tag{1}$$

where Q is the stream discharge (m^3/s) and V is the water volume stored within the beaver's pond (m^3). We estimated stream water discharge using a semi-distributed hydrological catchment model (PREVAH; Precipitation-Runoff-Evapotranspiration HRU Model; 28). We estimated pond water volume (V) by delineating the beaver pond boundary from aerial imagery and clipping the DEM to this polygon. With the QGIS's Volume Calculation tool we then quantified the volume enclosed within the pond footprint by integrating elevation differences across all DEM cells within the polygon. We validated this method by field surveys of six different beaver ponds.

4.5 Channel gradient

We estimated channel gradient between upstream and downstream locations by upstream and downstream elevations from the DEM using the "Extract values to point" tool. We then digitized polylines that follow the swisTLM3D [29] river line between each upstream-downstream pair and calculated their lengths (LL) with the QGIS field calculator. We computed slope as the elevation drop per horizontal distance using Equation 2:

$$\text{channel gradient}\,(\%) = \frac{\text{Upstream}_{altitude} - \text{Downstream}_{altitude}}{LL} \cdot 100 \tag{2}$$

4.6 Solar radiation

We computed the Solar radiation ($Wh\ m^{-2}$) of the stream reaches with the Grass r.sun.insoltime algorithm using the *qgisprocess* package in R [30]. This algorithm requires a digital surface model (DSM), a slope layer and an aspect layer. We calculated the DSM by combining a 2 m digital elevation model (DEM; 31) and a 1m digital model of vegetation height (resampled to the resolution of the DEM) [32]. Slope and aspect were computed from the 2 m DEM using the *terra* package in R [33].

4.7 Primary producers variables

We sampled primary producers' abundance, namely phytoplankton, macrophyte and soil litter cover at 16 sites (Figure 3a). To estimate phytoplankton abundance, we collected 50L of water from the main beaver pond and concentrated it to a 2 L sample using a 10 m mesh. In the laboratory, we counted and identified phytoplankton with a dual-magnification dark-field imaging microscope that captured all particles in the size range between ~10 m and ~1 cm using a 5.0× magnification objective. We estimated macrophyte abundance by counting each individual plant detected in the beaver ponds. We estimated soil litter percentage cover in a 1x5 m plot located at 0.5 m distance from the edge of the beaver pond in each site. We defined all dead plant material, such as leaves, but also twigs <7cm circumference, as soil litter.

4.8 Statistical analyses

4.8.1 Direction of the beaver-mediated DOC change (first research question)

To estimate the direction of the beaver-mediated DOC change we modelled the raw upstream and downstream DOC concentrations rather than their pre-computed difference, so that the analytical measurement error of each concentration (0.2 mg L^{-1}) is propagated exactly to ΔDOC. We fitted a Bayesian hierarchical measurement-error model in which the observed concentration of each sample is a noisy measurement of a latent true concentration (`mi(0.2)` in *brms*), and the latent concentrations follow a Student-*t* distribution (to accommodate the heavy-tailed distribution of DOC) whose mean is a season × location (upstream/downstream) factorial, with random intercepts for stream and for stream × season that pair the upstream and downstream samples of the same stream and sampling occasion [34]. The average ΔDOC of each season is derived from the posterior as the difference between the downstream and upstream season means, and the posterior predictive distribution of ΔDOC for a single sampling of a stream as that difference plus the difference of two residual draws. We considered a net change in DOC larger than 0.5 mg/L as ecologically relevant: when $\Delta DOC > 0.5$ mg/L, the pond is a DOC source within the ecosystem; when $\Delta DOC < -0.5$ mg/L, the pond is a DOC sink. We applied the Bayesian Region of Practical Equivalence (ROPE) to values between -0.5 *and* 0.5 mg/L; the ROPE defines an interval around zero where effects are too small to be of importance [35]. This ± 0.5 mg/L threshold is justified because changes of this magnitude are near levels at which DOC concentrations begin to affect light penetration, nutrient dynamics, and the balance between autotrophy and heterotrophy in freshwater ecosystems [36, 37]. We applied the ROPE (i) to the posterior of the seasonal average ΔDOC and (ii) to the posterior predictive distribution of ΔDOC for a single sampling (Figure 1a): the shares of the latter below, within and above the ROPE are the posterior predictive probabilities that one sampling of a beaver-engineered stream shows a sink, a negligible or a source response. These probabilities describe the variability of individual responses and are not percentages of streams; with two samplings per stream, stream-specific ΔDOC values cannot be resolved (Appendix S1: Figure S6).

4.8.2 Drivers of the beaver-mediated DOC change: structural causal model (second research question)

To answer the second research question, we used a two-step analytical approach combining causal modeling and Bayesian inference to integrate prior knowledge and account for uncertainty in our parameter estimates. First we described the set of possible data-generating processes that we hypothesized to be causing ΔDOC variation (i.e., Structural Causal Model, SCM; the hypothesized DAG is shown in Figure 1b and, with its thirteen causal paths numbered, in Appendix S1: Figure S1; each numbered path corresponds to one science-based hypothesis motivated in Appendix S1: Table S1) [38]. Second, we proceeded to infer the relationships between variables using a multivariate linear model within a Bayesian framework. A DAG makes the causal assumptions of an analysis explicit, and the identification of causal effects is conditional on those assumptions [38, 39]. Because different DAGs encode different hypotheses on how the system works, it is recommended to specify and test alternative DAGs and to report their estimates transparently, so that the robustness of the conclusions to the assumed causal structure can be assessed [40, 41]. We therefore also tested an alternative, extended SCM in which the lateral connectivity between the stream and adjacent wetlands influences water residence time, and hence ΔDOC (Appendix S1: Figures S8–S9). The full model specification is given below (Equation 4); the tested hypotheses for each causal path, the priors, residuals and posterior predictive checks are reported in Appendix S1 (Figures S2–S4). In the SCM, ΔDOC enters as the outcome with its propagated measurement error (Equations 3–4).

In this study we used the structural causal model (SCM) framework to answer the research questions (*sensu* 38). The SCM can be considered as a directed acyclic graph (DAG) with nodes and edges. The nodes indicate the relevant variables of the ecosystem and the edges indicate direct causal influences. The edges are acyclic because a cause is not at the same time an effect. We fitted the SCM with the Bayesian framework to accommodate the uncertainty inherent in ecological data and to incorporate prior distributions based on previous research and expert opinion [34].

We specified the SCM model as a system of linear equations, where each equation represents the relationship between a response variable and its predictors. We statistically defined the full Bayesian SCM as in Equation 4.

Given that the error in both the downstream and upstream DOC measurements was 0.2 $mg\ C\ L^{-1}$, we estimated the error of ΔDOC (i.e., E) by error propagation using Equation 3:

$$E = \sqrt{0.2^2 + 0.2^2} = \sqrt{0.08} \approx 0.283\ mg\ L^{-1} \tag{3}$$

Specifically, we treated the measurement error as missing data and imputed it using a normal distribution with a mean of 0 and a standard deviation of 0.283 mg/L. The overall model comprised a measurement model and an outcome multivariate model. First, we accounted for observational uncertainty by specifying a measurement model that links the observed $\Delta DOC_i^{obs/imputed}$ to a latent (true) DOC change, ΔDOC_i^{true}:

$$
\begin{aligned}
\textbf{Measurement model:}\quad \Delta\mathrm{DOC}_i^{\mathrm{obs}} &\sim \mathcal{N}(\Delta\mathrm{DOC}_i, 0.283^2) \\
\textbf{Outcome multivariate model:}\quad \Delta\mathrm{DOC}_i &\sim \text{Student-}t(\nu, \mu_i, \sigma_{\Delta\mathrm{DOC}}) \\
\mu_i &= \beta_1 \cdot \mathrm{DOC}_{\mathrm{upstream},i} \\
&\quad + \beta_2 \cdot \mathrm{Macrophytes}_i \\
&\quad + \beta_3 \cdot \mathrm{Phytoplankton}_i \\
&\quad + \beta_4 \cdot \mathrm{Litter\ Cover}_i \\
&\quad + \beta_5 \cdot \mathrm{WRT}_i \\
\textbf{Predictors multivariate model:}\quad \mathrm{Macrophytes}_i &\sim f(\mathrm{solar\ radiation}_i, \mathrm{WRT}_i, \dots) \\
\mathrm{Phytoplankton}_i &\sim f(\mathrm{Macrophytes}_i, \mathrm{WRT}_i) \\
\mathrm{Litter\ Cover}_i &\sim f(\mathrm{solar\ radiation}_i) \\
\mathrm{WRT}_i &\sim f(\mathrm{DamHeight}_i, \mathrm{NDams}_i, \dots) \\
&\vdots \\
\textbf{Priors:}\quad \alpha_{\mathrm{summer}} &\sim \mathcal{N}(0, 10) \\
\alpha_{\mathrm{winter}} &\sim \mathcal{N}(0, 7.5) \\
\beta_2, \beta_3, \beta_4, \beta_5 &\sim \mathcal{N}(0.05, 0.1) \\
\beta_6 &\sim \mathcal{N}(0, 0.2) \\
&\vdots \\
\sigma_{\Delta\mathrm{DOC}} &\sim \text{Student-}t(3, 0, 2.5) \\
\sigma_{\mathrm{upstream}} &\sim \mathcal{N}(0, 0.5) \\
\nu &\sim \mathrm{Gamma}(2, 0.1)
\end{aligned}
\tag{4}
$$

We used the mi() function from the R package *brms* (version 2.23) to account for the measurement error of 0.283 mg/L on the response variable ΔDOC. We imputed missing data during model fitting by treating them as additional parameters. We ensured that missing data in both response and predictor variables were modeled rather than excluded by using the mi() function. This function treats missing values as additional parameters and estimates them jointly with the model.

We generated 20,000 (four chains run for 10,000 iterations discarding the first 3000 as burn-in) Markov chain Monte Carlo (MCMC) samples from the posterior distribution. We visually assessed if the SCM was a reasonably good description of the data by posterior predictive check and residuals (Appendix S1: Figures S3 and S4). All MCMC chains indicated convergence through falling within the threshold (i.e., R-hat < 1.05) specified by Gelman and Rubin [42].

If posterior parameter distributions had more than 90% probability mass on either side of zero ($P(\beta > 0)$ or $P(\beta < 0)$), then we assumed that there was statistical evidence for the positive or negative effect of the predictor on the response variable.

We did all the data cleaning and statistical analysis within the R environment (version 4.6.1). We conducted data cleaning with the tidyverse (version 2.0.0) package and created figures using ggplot2 (version 4.0.3). We used the R packages *brms* (version 2.23), tidybayes (version 3.0.7), and bayesplot (version 1.15.0), with the CmdStan backend via cmdstanr (version 0.9.0), for Bayesian inference.

4.9 The beaver amplification factor (M)

We defined the beaver amplification factor (M; unitless), which compares the magnitude of the DOC change between upstream and downstream of the beaver-engineered reach with the magnitude of the seasonal DOC change of the same stream upstream of the reach, both expressed per unit area (Figure 3b). The "floodplain-scale" reference is therefore the seasonal (summer minus winter) change in upstream DOC concentration normalised by the floodplain area of the catchment draining to the site: it is a reference for DOC variation generated at the valley-floor scale without beaver engineering, not a measured flux of DOC from the floodplain. Because M is a ratio of absolute values, it measures how much DOC changes per unit area irrespective of the sign of the change, and is thus complementary to ΔDOC, which measures its direction. The delineation of the catchment and floodplain area of each site is described below. We estimated the DOC change in the upstream floodplain area ($D^{seasonal}_{floodplain}$; $mg\ L^{-1}m^{-2}$) as in Equation 5:

$$D^{seasonal}_{floodplain} = \frac{DOC^{summer}_{upstream} - DOC^{winter}_{upstream}}{A_{floodplain}} \tag{5}$$

We estimated the DOC change along the beaver-engineered sites (D^{season}_{beaver}; $mg\ L^{-1}m^{-2}$) as in Equation 6:

$$D^{season}_{beaver} = \frac{DOC^{season}_{down} - DOC^{season}_{up}}{A_{beaver}} \tag{6}$$

We defined the beaver amplification factor (M) as in Equation 7:

$$M = \frac{|D^{season}_{beaver}|}{\left|D^{seasonal}_{floodplain}\right|} \tag{7}$$

and its logarithm with Equation 8:

$$Y_{logM} = \log_{10}\left(|D^{season}_{beaver}| + c\right) - \log_{10}\left(\left|D^{seasonal}_{floodplain}\right| + c\right). \tag{8}$$

where c prevents $\log_{10}(0)$.

We defined a linear hierarchical model to estimate the average of Y_{logM}, allowing the mean to vary by season and accounting for site-level heterogeneity:

$$Y_{logM} \sim \mu_{logM,season} + u_{site} + \epsilon \tag{9}$$

where u_{site} represents site-specific deviations and ϵ the residual error of the beaver amplification factor (M). The coefficients $\mu_{logM,season}$ represent the average orders-of-magnitude difference between beaver and floodplain-scale DOC change in summer and winter. For example, $\mu = 5$ indicates the beaver-mediated DOC change is, on average, 10^5 times greater per unit area than the seasonal floodplain-scale DOC change, $\mu = 0$ indicates equal magnitude, and $\mu = -2$ indicates the beaver DOC change is 1/100 of the seasonal floodplain-scale DOC change. With this approach, we were able to directly compare beaver-mediated DOC changes with seasonal floodplain DOC variability and quantify how large the beaver-mediated DOC change is relative to the floodplain-scale one.

4.9.1 Definition of catchment and floodplain area

We delineated the catchment area by automatically assigning pour points to stream outlets at each site and entering these, along with a D8 pointer raster based on a 25 m DEM [43] into the *Watershed* algorithm in Whitebox [44] executed using the *Whitebox* package in R [45]. We calculated the floodplain area of each site using a custom-made function in R based on the elevation differential of the local terrain and the nearest channel, adapted from a method developed by **(author?)** [46].

Data and code availability. We confirm that all codes for data processing, including the processed data, main analyses, and supplemental analyses, are available on the GitHub repository https://github.com/leomarameo7/beavers_DOC.

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Conflict of interest statement. We declare no conflict of interest.

Supplementary information. Appendix S1 (Table S1, Figures S1–S9) is provided as a separate file: `supplementary/appendix_S1.pdf`.

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